# 1. Clinical Question & Core Hypothesis

## Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

## Working Hypothesis

PROS1 expression in breast tumor cells is positively associated with MKI67 expression (cellular proliferation), and this PROS1–MKI67 relationship is significantly moderated by stromal-to-tumor ratio (TSR). Specifically, in tumors with low TSR, high PROS1 confers a stronger proliferative signal (higher MKI67) than in high-TSR tumors. Consequently, PROS1-high, low-TSR tumors will have shorter progression-free interval (PFI) compared with PROS1-low or high-TSR tumors, indicating a PROS1–TSR interaction that provides prognostic information beyond PROS1 or TSR alone.

| Element | Value |
|---|---|
| Imaging feature | Stromal-to-tumor ratio (TSR) |
| Anchor gene(s) | PROS1, MKI67 |
| Primary endpoint | PFI |

# 2. Why This Gene — Not Others

## Primary hypothesis gene(s)

**PROS1:** PROS1 to proliferation (Ki-67), raising the question of when PROS1-associated proliferation translates into earlier progression.

**MKI67:** Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

# 3. Clinically Testable Predictions

**Prediction 1:** In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

# 4. Minimum Viable Validation Plan

## A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

## B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

### C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

## 5. Strengths, Caveats & Alternative Directions

### Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

### Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

### Alternative Directions